

An Analysis of NBA Player Salary Variation Across Different Player Statistics

The Significant Others - Siri, Marin, ReNata

Introduction

The National Basketball Association has existed for around 80 years and consists of 30 teams across North America, featuring some of the best basketball players in the world. (NBA, 2019) Within the league, player salaries vary substantially and are influenced by both individual performance and league/team structure (Li, 2014) In addition, the league has a salary cap system in which there are limits on the total amount teams can spend on player salaries. (Sigler, 2018) This cap is determined by the NBA’s total income; however, it is also flexible as there are ways that teams can go over the limit. (NBA, 2025)

It is important to understand what factors have the biggest impact on player salaries. Thus, in this analysis, we applied multiple linear regression to analyze how factors such as playing time, usage percentage, age group, and conference relate to salary variation among players. Furthermore, our research was guided by the following questions:

- 1) Do NBA players in the Western Conference earn higher average salaries compared to players in the Eastern Conference?
- 2) Do NBA players who average more minutes of playing time per game in a season earn higher salaries than players who average fewer minutes per game?
- 3) Are NBA player salaries higher for the “old” age group compared to the other age groups (prime, young, veteran)?

Each of these questions is motivated by different potential factors that impact salary variation. Specifically, conference affiliation can reflect differences such as competition. Playing time serves as a measure for player importance and contribution, which may contribute to higher or lower salaries. Additionally, age helps show the relationship between player performance versus experience, representing how salaries may be reflective of both athletic ability and veteran status.

Ultimately, this analysis focuses specifically on how performance related and demographic factors influence NBA player salaries, providing insight into what drives variation in salaries.

Data Summary

Metadata Description

Two primary sources were merged for this analysis. One consisted of player salaries from the 2024-2025 season and the other included player statistics from the 2023-2024 season. The

source providing salary information for the players came from the official ESPN website, while the other dataset containing player statistics was taken from the advanced stats section of the official NBA website for 2023-2024 season. Neither source provided a specific publication date or a specified author. However, both are widely reputable platforms for sports related data.

We manipulated several of our variables to better support our analysis. Player salaries were converted into millions of dollars by dividing the original values by 1,000,000. Net Rating (NETRTG) was converted into a qualitative variable with three levels: low impact, average, and high impact. These groupings were based on 33rd and 66th percentile cutoffs. Age was manipulated to be a qualitative variable. Players were categorized as “Young” (18-23), “Prime”(24-28), “Veteran”(29-34), or “Old” (35+). Additionally, we created a conference variable, categorizing players as either “East” or “West” based on their team affiliation.

EDA Summary

Response Variable (Salary): The distribution of NBA player salaries for the season 2024-2025 ranges between 20 and 60 million dollars. The distribution indicates an overall downward trend(skewed right), with most players earning between 20 and 30 million. The distribution is also unimodal and there are no extreme outliers.

Quantitative EDA Summary:

The Player Impact Estimate seems to have the strongest correlation with a player’s salary the at 0.59. True shooting percentage, minutes played/game,and usage percentage have lower correlations with a player’s salary of 0.49, 0.45, and 0.47 respectively. The weakest correlation is between offensive rating and player salary at 0.36.

Qualitative EDA Summary:

There appears to be a relationship between age group and player salary because the mean value of y is different at each level. Players that belonged to the “Old” age group in the previous season earned the highest salary (44.67), while young and prime players have relatively similar average earnings (33.59 and 32.46), and veteran players earned an average of 36.57.

There appears to be a relationship with conference and player salary because the mean value of y is different for each level. Players in the Eastern conference had an salary of 32.70, while the Western conference players had a higher average salary of 36.49.

There appears to be a relationship between the net rating tier of a player and their salary

because the mean value of y increases across levels. Low impact players have the lowest mean salary of 29.01, while average players have an average salary of 36.05, and high impact players have the highest average salary of 38.89.

Interactions Summary:

The interaction plot shows that there may be an interaction between net rating tier and age group on salary. The slopes do not appear to be parallel and there is some crossing over that occurs, indicating that the effect of net rating tier on salary may be dependent on age.

The grouped scatterplot indicates a generally positive correlation between player impact estimate and salary. The effect of a player's impact estimate on salary does not seem to be dependent on age as the slopes are generally similar.

There may be an interaction between the average playing time/game (MIN) and net rating tier. The slopes are fairly different as the high impact slope is the steepest, meaning that the highest impact players who play for a longer period of time per game earn the highest salaries. Because the lines are not parallel, it serves as evidence of a potential interaction.

Analysis and Results

We fit an initial model with our quantitative variables of interest (MIN, OFFRTG, TS_PCT, USG_PCT, PIE) to assess multicollinearity. Looking at the correlation plot, there is a concern of moderately strong pairwise relationships with:

- USG_PCT & MIN ($r=0.56$)
- PIE & TS_PCT ($r=0.51$)
- USG_PCT & PIE ($r=0.65$)

After checking the VIF values, there is no evidence of severe multicollinearity because none of the individual VIF values are greater than 10 and the average VIF is not greater than 3. Doing variable screening using stepwise regression with thresholds of $\text{pent}=0.15$ and $\text{prem}=0.15$ allowed us to narrow down our quantitative variables to MIN, TS_PCT, and USG_PCT.

From here, we proceeded to model fitting and fit our first model with the three quantitative variables that remained after variable screening. We did not have any quantitative x quantitative interactions to add at this time. The global F test resulted in a p value of

0.0000000000000331, allowing us to reject the null hypothesis and conclude that the model with MIN, TS_PCT, and USG_PCT is adequate at predicting the salary of NBA players in millions of dollars. Then, the individual T tests revealed that all three of our variables (MIN, TS_PCT, and USG_PCT) were significant with p-values of 0.0112, 0.00000000145, and 0.0000765 respectively, meaning that these variables are significant predictors of NBA player salary. We then fit an updated model, adding in our qualitative variables of interest (AGE_GROUP, CONFERENCE, and NETRTG_TIER). We performed nested F tests on the age group and net rating tier variables, which resulted in p-values of 0.380 and 0.05 respectively, meaning that both of these variables had to be removed from our model. Additionally, we performed an individual T test on conference, which resulted in a p-value of 0.278, leading us to remove this variable from our model as well. From here, we intended to add in our quantitative x qualitative interaction between MIN and NETRTG_TIER, but did not proceed because one of the main effects (net rating tier) was dropped. Therefore, after this phase of model building, our model consisted of just our three quantitative variables - MIN, TS_PCT, and USG_PCT.

To check the model assumptions, we looked at residual plots, residual vs fitted plots, a histogram of the residuals, and a QQ plot. Generally speaking, our individual residual plots didn't show any clear trends, but there was one noticeable outlier on the plot for MIN. Overall, we do not believe that our mean zero (lack of fit) assumption was violated. Our residual plots and residual vs fitted plots also did not show any clear fanning patterns. For the MIN variable, there was a slight fanning pattern, but after trying to correct it through transformations and not seeing much of a change, we decided to keep it as is. Again, generally speaking, we do not believe that the constant variance (homoscedasticity) was violated. However, the points on the QQ plot did not fall linearly, with many points above or below the line and the histogram was also slightly left skewed. Therefore, our normality assumption was violated, but with a large enough sample size, violations against normality can be considered robust for MLR. Additionally, we plan to use the box-cox method to address this violation. Finally, we do not believe that the statistics of one player depend on another player, so the independence violation is likely not violated.

We then conducted an analysis of influential observations using leverage, studentized residuals, Cook's Distance, and deleted studentized residuals. Observations 10, 23, 42, and 50 were identified as high leverage points based on the threshold $2(3+1)/78 = 0.1026$. Studentized residual analysis identified observations 7, 11, 18, 24, 37, 39, and 44 as potential outliers in the response variable due to values exceeding ± 2 . Cook's Distance identified observations 11, 18, 24, 37, 39, and 42 as influential with the threshold $4/78$. Deleted stu-

dentized residuals confirmed a similar set of influential observations (7, 11, 18, 24, 37, 39, and 44). However, there were no observations that simultaneously exceeded thresholds for leverage, Cook's distance, studentized residuals, and standardized residuals. Therefore, from here, we went back into our data and looked at consistently repeated observations such as 7,11,18,24,37,39,and 44 to see if any of the values for the quantitative variables seemed extremely high or low, but did not see anything unusual as the values for the variables (MIN, TS_PCT, and USG_PCT) for these specific players fell within the range of the remaining players.

After looking at dffits and dfbetas, it seemed like observation 37 was the most influential overall, with the largest DFFITS value (0.69). In addition, DFBetas show that removing observation 37 would affect the coefficients by a fairly large amount for MIN (0.543) and USG_PCT (-0.519). Observation 11 is also highly influential, with the second largest DFFITS value (-0.59), and DFBetas indicating a noticeable effect on TS_PCT (-0.406). Finally, to determine whether observations 11 and/or 37 should be removed, we refit three separate models to determine whether it would be best to remove both observations, only remove 37, only remove 11, or keep both of them.

Ultimately, we decided to not remove either of these observations. Although the model excluding both observations had the highest adjusted R squared value (0.5934), the improvement over the other models was relatively small and not enough to justify removing these data points. Specifically, the adjusted R squared values for the original model, the model excluding only observation 11, and the model excluding only observation 37 were 0.5424, 0.567, and 0.5661, respectively.

However, because the normality assumption was initially violated, we applied a box-cox transformation to the response variable to see if the violation can be addressed. After performing the box-cox transformation, a log transformation transformation was applied to the response variable (SALARY), and the normality assumption improved significantly, with most of the residuals falling close to the line on the QQ-plot. Additionally, while the adjusted R-squared valued remained fairly similar at around 0.54, the residual standard error for the transformed model improved significantly. Prior to the box-cox transformation, the RSE was 6.381, while the residual standard error for the transformed model is 1.22 after taking the inverse to properly scale the units. Therefore, we decided to apply the box-cox transformation.

Our final model includes the three quantitative variables - MIN, TS_PCT, and USG_PCT, keeps all observations, and transforms the response variable (SALARY). The global F-test shows that our model is adequate with a p-value of 0.0000000000002866. Each individual

predictor is also significant as p-values for each of them are significantly below 0.05.

Conclusions

The prediction equation is given by: $\widehat{SALARY} = -0.6948 + 0.02222MIN + 0.05210TS_PCT + 0.02139USG_PCT$

For each additional minute of playing time, the expected log of salary increases by 0.022 million dollars, holding true shooting percentage and usage percentage constant. Additionally, for a one unit increase in the true shooting percentage, the expected log of salary increases by 0.052 million dollars, holding MIN and usage percentage constant. Finally, for a one unit increase in usage percentage, the expected log of salary increases by 0.021 million dollars, holding MIN and true shooting percentage constant.

To exemplify the model being used for prediction, consider a player with 33.6 minutes of playing time/game, a true shooting percentage of 64.4, and a usage percentage of 38.7. According to the model, we are 95% confident that a NBA player who plays an average of 33.6 minutes, has a true shooting percentage of 64.4 and a usage percentage of 38.7 will have a salary between \$44.863 and \$106.265 million. Additionally, the player's estimated salary is 69.045 million. The actual observed salary for this player is \$51.416 million, meaning that the model overestimates by approximately \$17 million.

If minutes played increase to 43.6 while holding all other variables constant, the predicted salary increases to approximately \$86.227 million, with a 95% prediction interval of \$55.89 million to \$133.03 million. This demonstrates that increased playing time could be correlated with higher predicted salary. However, with respect to our other proposed research questions, variables such as conference and age group were found to be insignificant through the model building process and were removed from the model. As a result, conclusions regarding their relationship with salary cannot be determined in this analysis.

There are also several limitations to our analysis process that are important to consider. First, many variables were dropped throughout the model building process that could potentially be important in determining a player's salary such as age, conference, and player impact estimate. This limits the model's ability to fully explain the variation in NBA salaries. Additionally, NBA salaries are also dependent on external factors such as the salary cap system and contract structure, which are not accounted for in this model. Moreover, while the box-cox transformation improved the normality assumption, it also reduces the interpretability of the model.

In the future, it would be beneficial to add additional variables such as points per game and contract year, as well as increasing the sample size and consider using data across multiple seasons. These improvements could potentially contribute to enhancing our model.

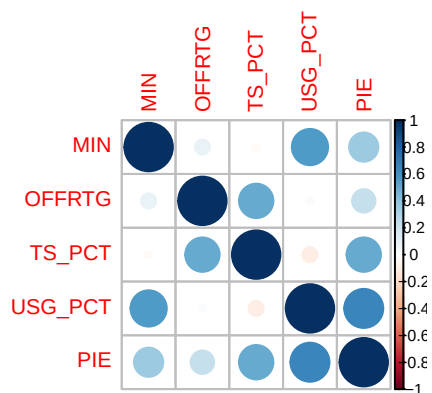
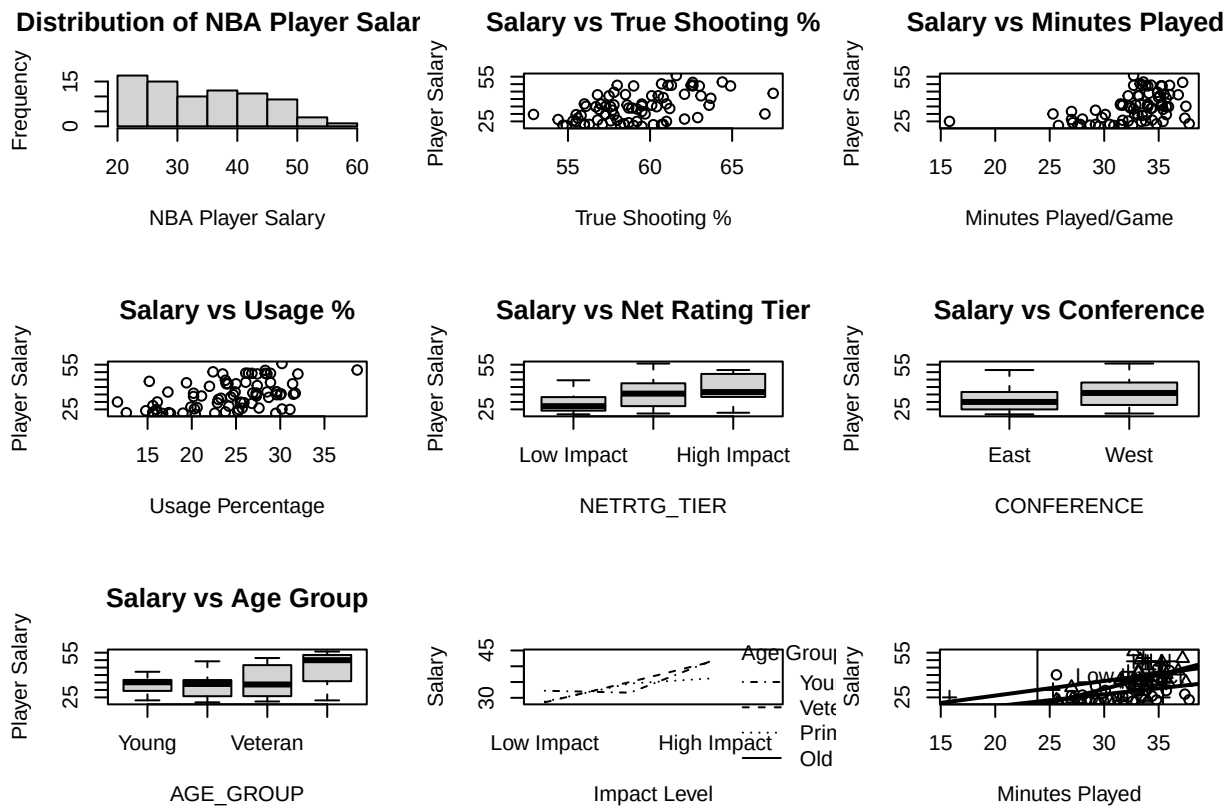
Appendix A: Data Dictionary

Variable Name	Abbreviated Name	Description
Salary	SALARY	Player salary in millions of dollars
Player Impact Estimate	PIE	Player's overall contribution to the game; expressed as a percentage
True Shooting Percentage	TS_PCT	Measure player's efficiency at scoring (percentage)
Minutes Played	MIN	Average minutes played per game
Offensive Rating	OFFRTG	Estimates offensive efficiency by calculating points scored per 100 possessions
Usage Percentage	USG_PCT	Estimates the percentage of a team's possessions a player uses while on the court (percentage of possessions)
Net Rating Tier	NETRTG_TIER	Derived from quantitative variable (NETRTG); classifies players into three groups: low impact, average, or high impact
Conference	CONFERENCE	Indicates whether a player is in the Eastern or Western conference
Age Group	AGE_GROUP	Grouping of players into young, prime, veteran, or old based on age

Appendix B: Data Rows

	PLAYER	TEAM	AGE	GP	Wins	Losses	MIN	OFFRTG	DEFRTG	NETRTG	ASST_PCT
1	Kevin Durant	PHX	35	75	44	31	37.2	118.1	113.3	4.8	21.3
2	Kawhi Leonard	LAC	33	68	44	24	34.3	119.3	111.1	8.2	16.5
3	Jayson Tatum	BOS	26	74	57	17	35.7	121.7	110.5	11.2	21.1
4	Devin Booker	PHX	27	68	43	25	36.0	119.3	113.5	5.8	29.9
5	Michael Porter Jr.	DEN	26	81	57	24	31.7	120.2	112.1	8.1	6.4
6	Paul George	LAC	34	74	48	26	33.8	121.2	112.1	9.1	15.7
	AST_TO	AST_RATIO	OREB_PCT	DREB_PCT	REB_PCT	TO_RATIO	EFG_PCT	TS_PCT	USG_PCT		
1	1.55	16.9	1.6	15.5	8.9	10.9	58.1	62.6	28.3		
2	2.05	14.9	3.9	13.8	9.1	7.3	58.5	62.6	26.1		
3	1.94	16.7	2.6	18.7	11.0	8.6	55.2	60.4	29.6		
4	2.65	22.1	2.4	9.8	6.3	8.3	55.0	61.1	28.9		
5	1.33	9.1	4.2	17.7	11.1	6.9	58.6	60.1	20.2		
6	1.71	14.8	1.6	13.8	8.0	8.7	56.9	61.3	26.4		
	PACE	PIE	POSS	RK	Position	SALARY	AGE_GROUP	CONFERENCE	NETRTG_TIER		
1	100.24	14.8	5832	4	F	51.17902	Old	West	Average		
2	97.77	15.4	4747	6	F	49.35000	Veteran	West	High Impact		
3	98.64	15.5	5436	40	F	34.84834	Prime	East	High Impact		
4	99.96	14.2	5103	7	G	49.20580	Prime	West	High Impact		
5	97.81	10.6	5247	34	F	35.85995	Prime	West	High Impact		
6	97.47	12.6	5080	8	F	49.20580	Veteran	West	High Impact		
	SALARY_bc										
1	4.095991										
2	4.056603										
3	3.681477										
4	4.053437										
5	3.712229										
6	4.053437										

Appendix C: Exploratory Data Analysis



Appendix D: Final Model Output and Plots

Call:

```
lm(formula = SALARY_bc ~ MIN + TS_PCT + USG_PCT, data = nba_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.51714	-0.13441	0.00039	0.13899	0.40689

Coefficients:

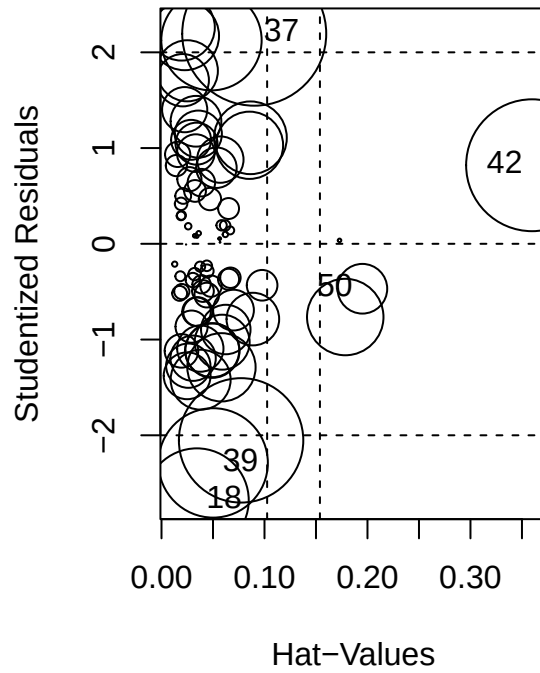
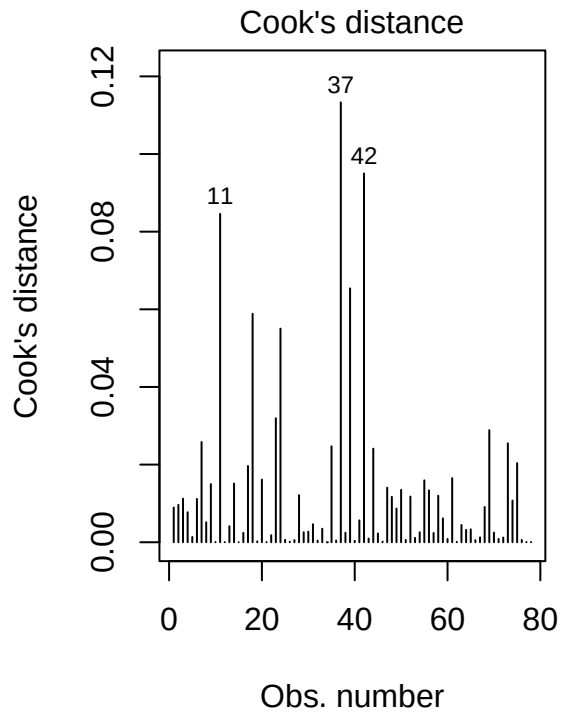
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.694759	0.509576	-1.363	0.17689
MIN	0.022221	0.007909	2.810	0.00634 **
TS_PCT	0.052096	0.007756	6.717	0.00000000329 ***
USG_PCT	0.021394	0.005101	4.194	0.00007512406 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

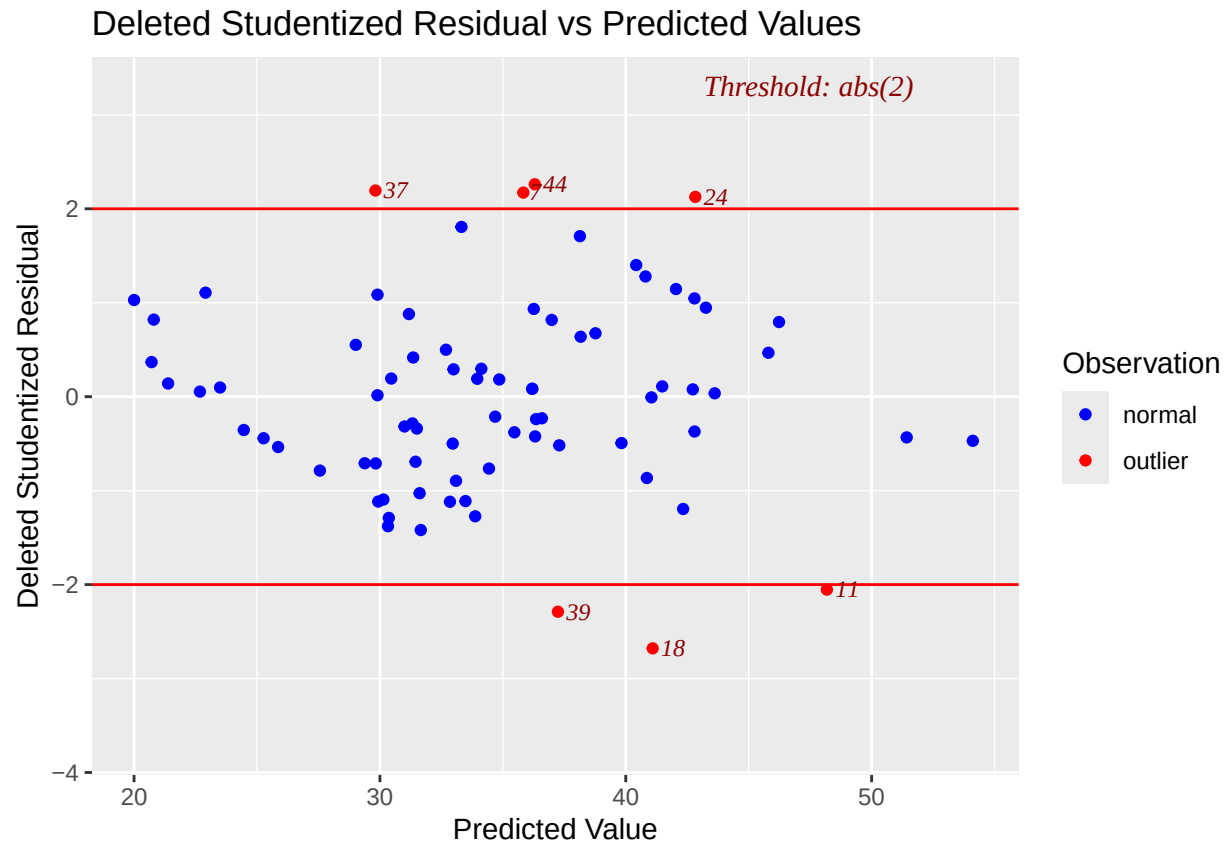
Residual standard error: 0.1979 on 74 degrees of freedom

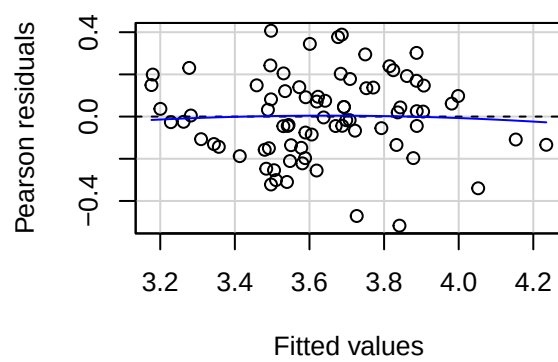
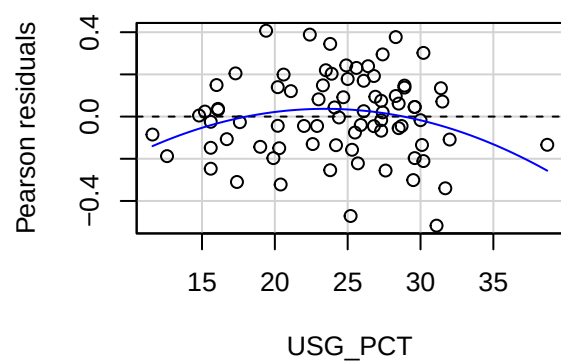
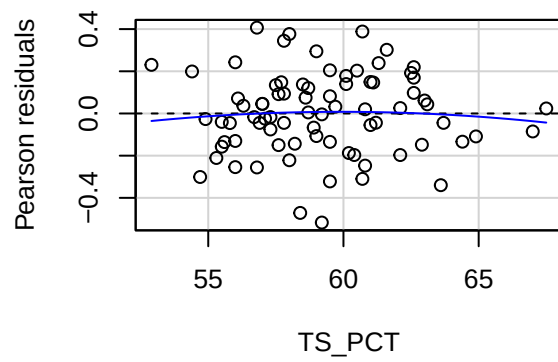
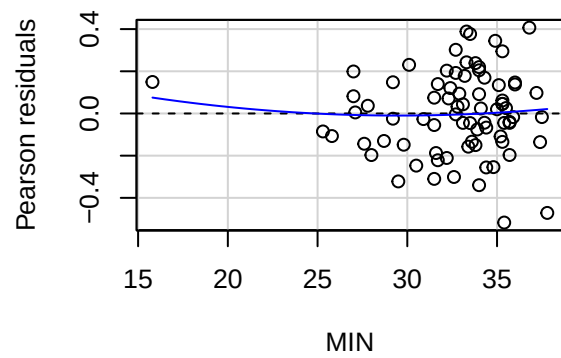
Multiple R-squared: 0.5619, Adjusted R-squared: 0.5442

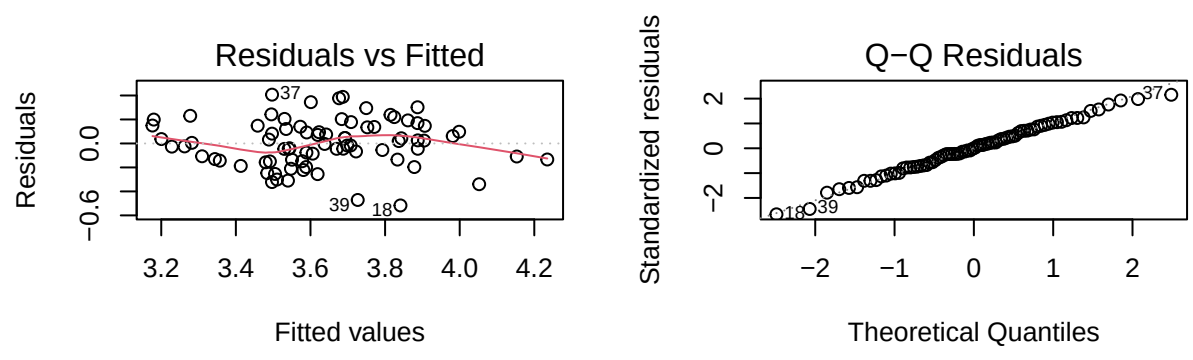
F-statistic: 31.64 on 3 and 74 DF, p-value: 0.0000000000002866



	StudRes	Hat	CookD
18	-2.6784581	0.03432816	0.05884751
37	2.1946989	0.09004397	0.11331386
39	-2.2886636	0.05015255	0.06539691
42	0.8204843	0.35982068	0.09501393
50	-0.4693516	0.19505435	0.01348733







Appendix E: References

Background

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Data Sources

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Players Advanced | Stats | NBA.com. (2025). NBA. <https://www.nba.com/stats/players/advanced?sort=W&dir=-1&Season=2023-24&SeasonType=Regular+Season>

Additional Help

Box-cox transformation. (n.d.). Wwww.css.cornell.edu. https://www.css.cornell.edu/faculty/dgr2/_static/files/R_html/Transformations.html